

3.6 Trigonometric Functions

Trigonometry is the branch of mathematics concerned with the relationships between angles and lengths in triangles. Trigonometric functions are widely used throughout engineering, physics, surveying, navigation, and signal analysis.

3.6.1 The Trigonometric Functions

Trigonometric functions take an angle as their input. Throughout this section we shall work in **degrees**. There are three primary trigonometric functions:

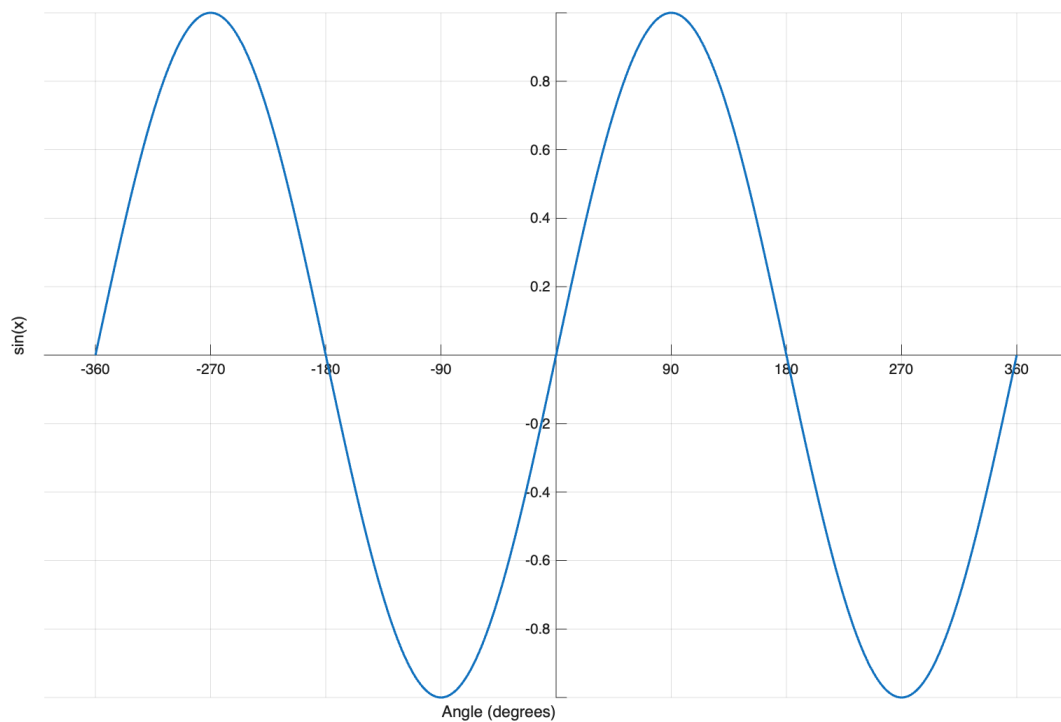
$$y = \sin(\theta)$$

$$y = \cos(\theta)$$

$$y = \tan(\theta)$$

3.6.1.1 The Sine Function

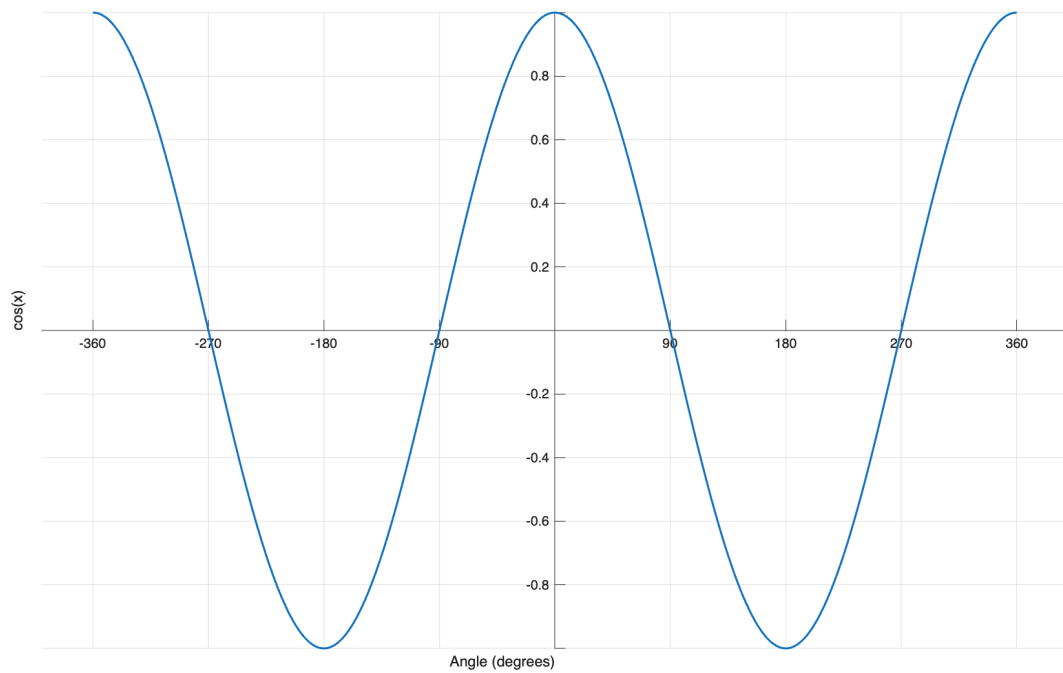
The sine function varies smoothly between -1 and 1 .



The graph repeats every 360° .

3.6.1.2 The Cosine Function

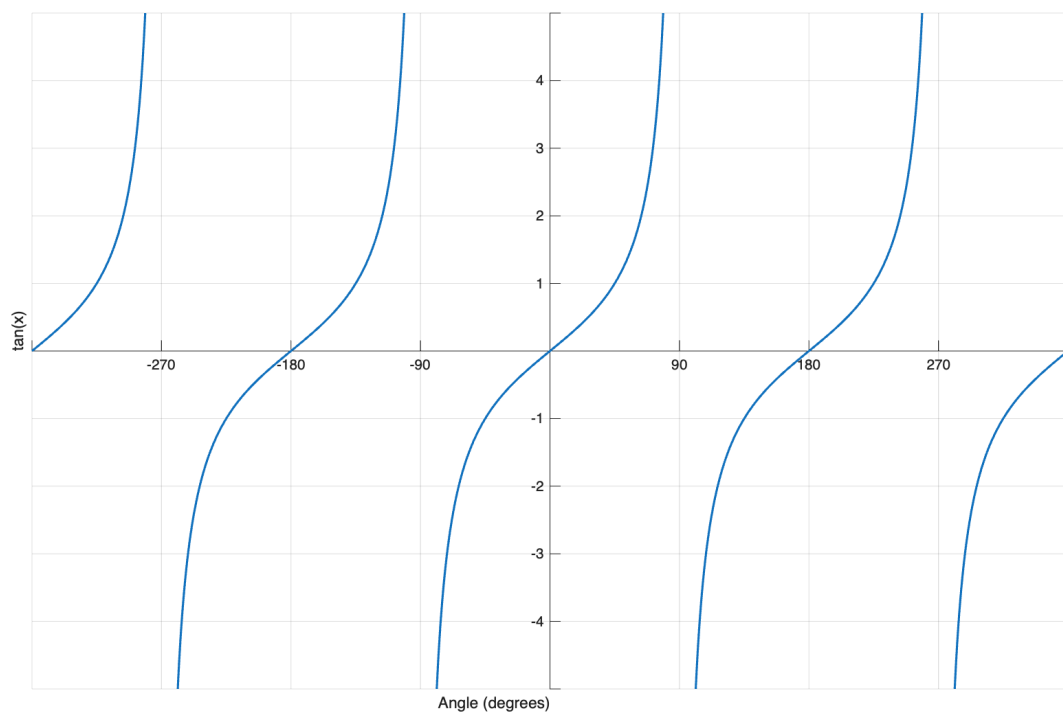
The cosine function also varies between -1 and 1 .



The graph repeats every 360° .

3.6.1.3 The Tangent Function

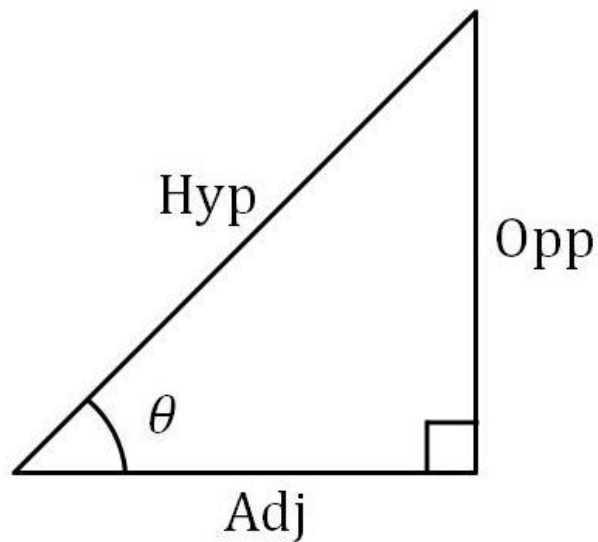
Unlike sine and cosine, the tangent function does not remain between fixed limits.



The function is undefined at 90° and 270° . The graph repeats every 180° .

3.6.2 Right-Angled Triangles

The trigonometric ratios allow unknown sides and angles in right-angled triangles to be calculated.



The three ratios are often remembered using:

SOH-CAH-TOA

Definition

For a right-angled triangle:

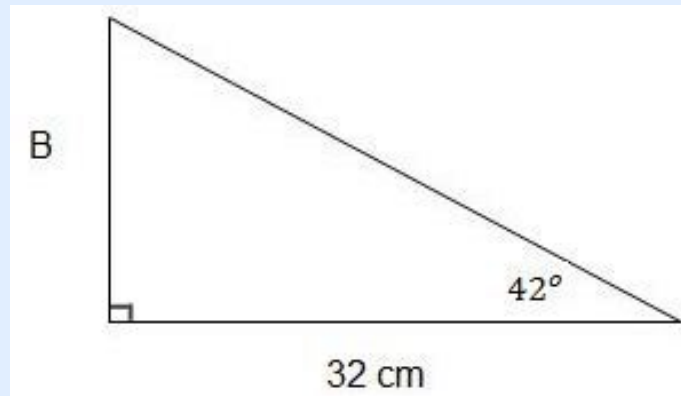
$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

Worked Example

Calculate the length B .



The opposite and adjacent sides are known, therefore we use the tangent ratio.

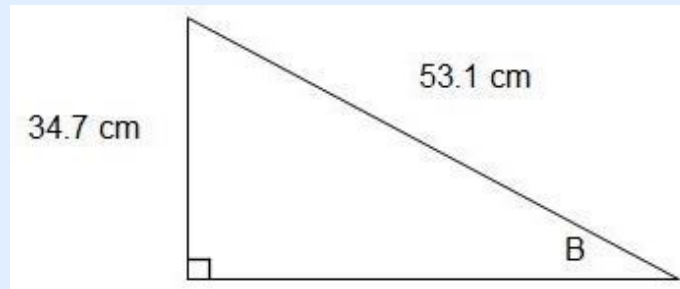
$$\tan(42^\circ) = \frac{B}{32}$$

$$B = 32 \tan(42^\circ)$$

$$= 28.81\text{ cm}$$

Worked Example

Calculate the angle B .



The opposite side and hypotenuse are known, therefore we use the sine ratio.

$$\sin(B) = \frac{34.7}{53.1}$$

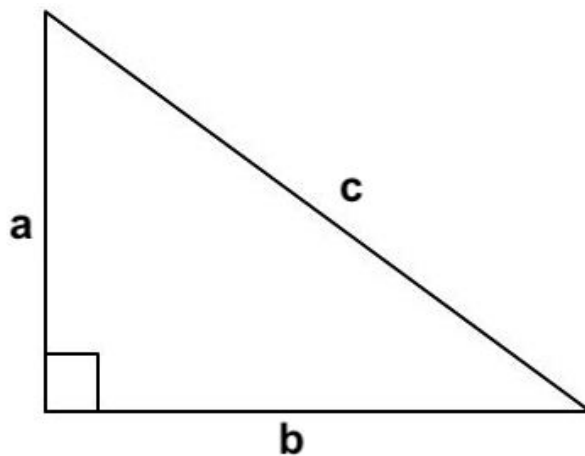
$$\sin(B) = 0.6535$$

Taking the inverse sine:

$$\begin{aligned} B &= \sin^{-1}(0.6535) \\ &= 40.80^\circ \end{aligned}$$

3.6.3 Pythagoras' Theorem

In a right-angled triangle, if two sides are known, the third side can be calculated using Pythagoras' theorem.



Definition

For a right-angled triangle:

$$a^2 + b^2 = c^2$$

where c is the hypotenuse.

Worked Example

A right-angled triangle has sides of length 5 and 12. Calculate the hypotenuse.

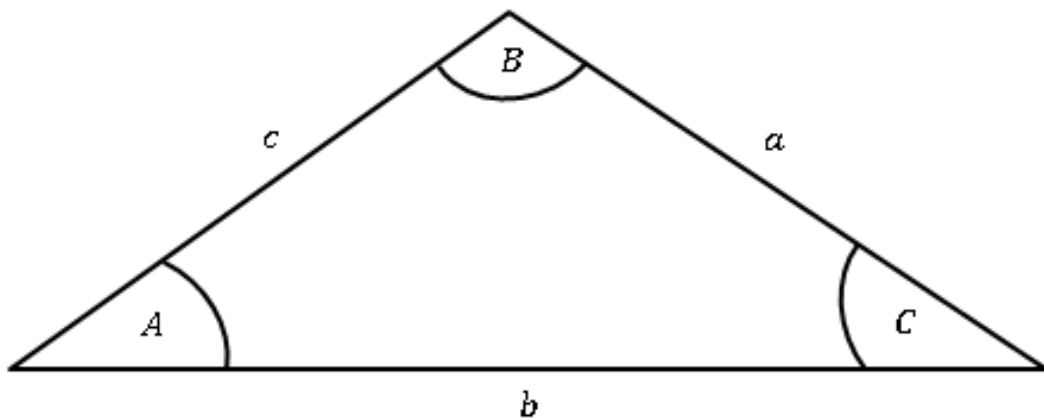
$$\begin{aligned}c^2 &= 5^2 + 12^2 \\&= 25 + 144 \\&= 169\end{aligned}$$

Therefore,

$$\begin{aligned}c &= \sqrt{169} \\&= 13\end{aligned}$$

3.6.4 Non-Right-Angled Triangles

For non-right-angled triangles, the trigonometric ratios above are insufficient. Instead, we use either:



- the sine rule,
- the cosine rule.

3.6.4.1 The Sine Rule

Definition

The sine rule states that

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

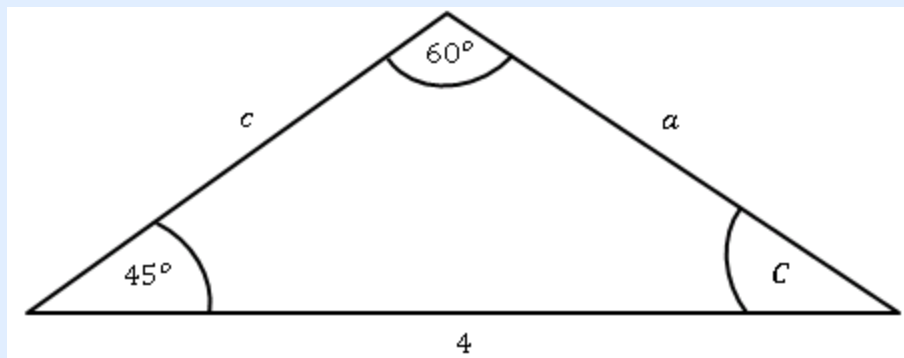
This rule is useful when:

- two angles and one side are known, or
- two sides and a non-included angle are known.

Worked Example

A triangle has

$$A = 45^\circ, \quad B = 60^\circ, \quad b = 4$$



Calculate side a . Using the sine rule:

$$\frac{a}{\sin(45^\circ)} = \frac{4}{\sin(60^\circ)}$$

Therefore,

$$\begin{aligned} a &= \frac{4 \sin(45^\circ)}{\sin(60^\circ)} \\ &= 3.27 \end{aligned}$$

3.6.4.2 The Cosine Rule

Definition

The cosine rule states that

$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

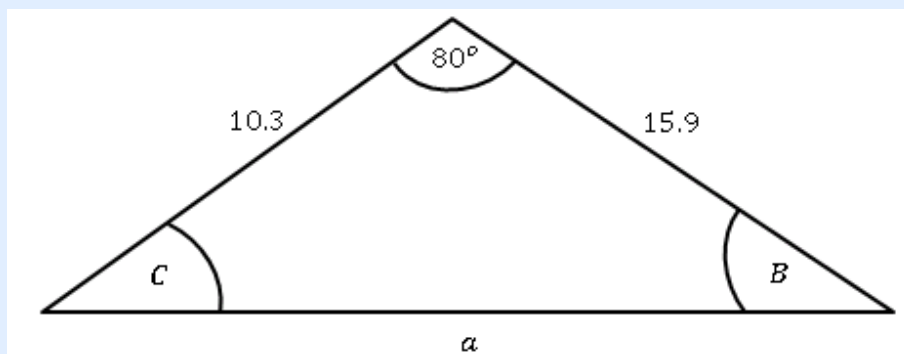
This rule is useful when:

- all three sides are known, or
- two sides and the included angle are known.

Worked Example

Given

$$b = 10.3, \quad c = 15.9, \quad A = 80^\circ,$$



determine a . Using the cosine rule:

$$\begin{aligned} a^2 &= 10.3^2 + 15.9^2 - 2(10.3)(15.9) \cos(80^\circ) \\ &= 302.02 \end{aligned}$$

Therefore,

$$\begin{aligned} a &= \sqrt{302.02} \\ &= 17.38 \end{aligned}$$

3.6.5 Applications of Trigonometry

Engineering Application

Trigonometry is used to determine distances and heights that cannot easily be measured directly. Common applications include:

- measuring the height of buildings,
- determining the distance to offshore structures,
- navigation and positioning systems.

Engineering Application

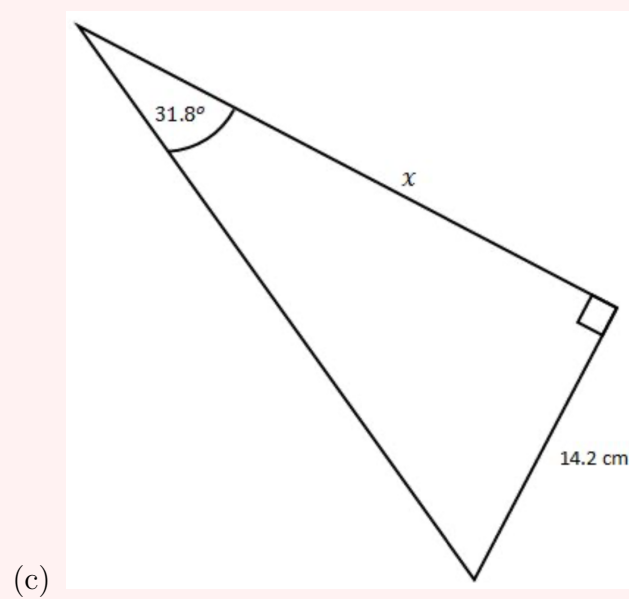
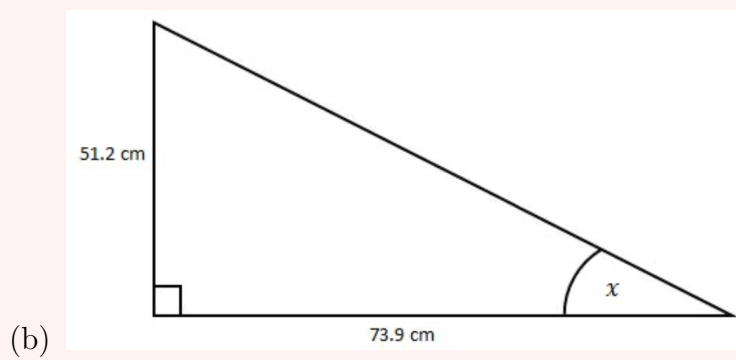
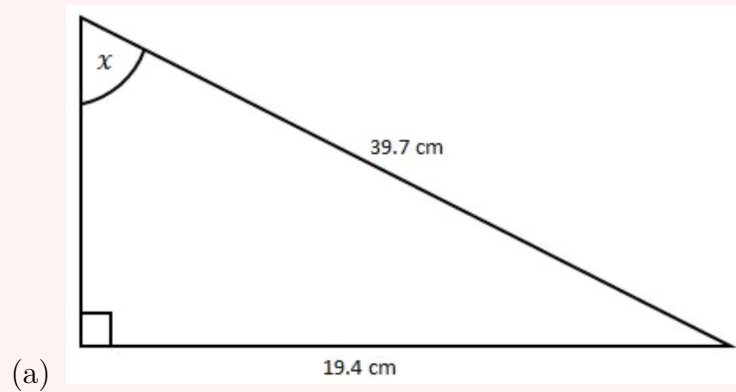
Engineers use trigonometry when analysing structures, forces, mechanisms and components that contain angled members. Examples include:

- truss structures,
- cranes,
- bridges,
- mechanical linkages.

3.6.6 Exercises

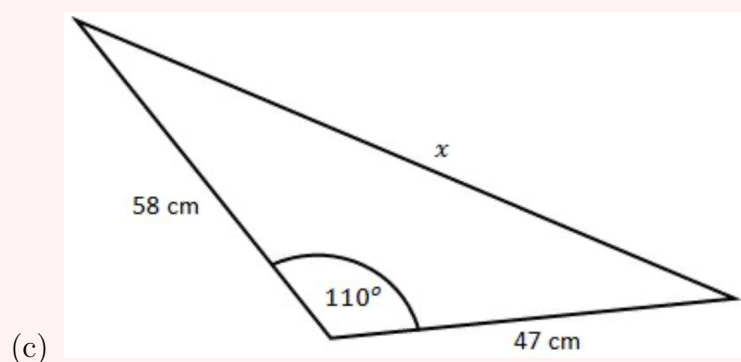
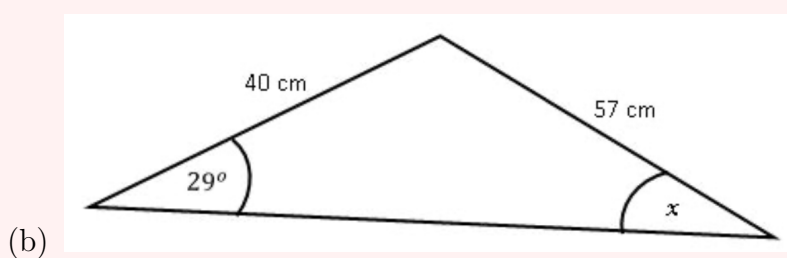
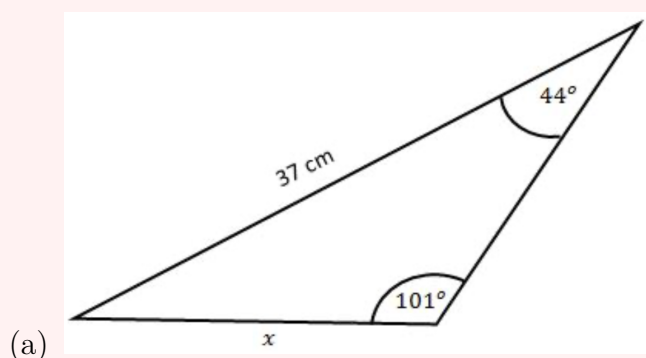
Exercise

1. Calculate the value of x :



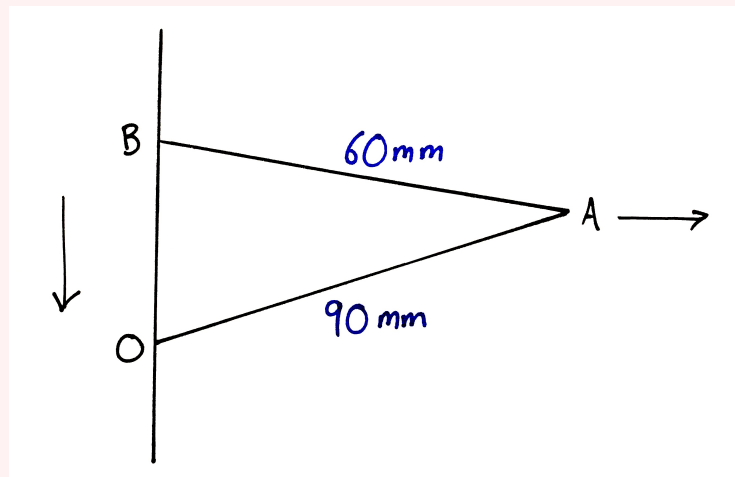
Exercise

2. The angle of elevation of the top of a tree from a point on the ground that lies 10m from the base of the tree is 28° . What is the height of the tree, to 1 decimal place?
3. The angle of elevation of the top of a 20m high mast from a point at ground level is 34° . How far is the point from the foot of the mast? Give your answer to 2 decimal places.
4. A supporting cable of length 30m is fastened to the top of a 20m high mast. What angle does the cable make with the ground? How far away from the foot of the mast is it anchored to the ground?
5. Calculate the value of x :



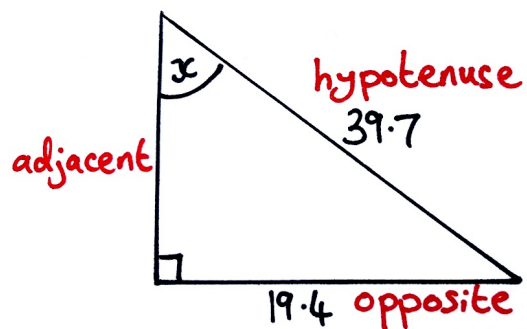
Exercise

6. What is the height (to 1 decimal place) of an isosceles triangle with base angle 65° and sloping sides with length 10cm? What is the length of the base of this triangle (to 1 decimal place)?
7. Two radar stations located 20 miles apart each detect an aircraft situated between them. The angle of elevation measured by the first station is 35° , whereas the angle of elevation measured by the second station is 15° . Determine the altitude of the aircraft.
8. The figure shows part of a hinge for a window. As the window is opened, part B moves down to the fixed point O while A moves to the right. When angle $A = 22^\circ$, determine the length OB.



3.6 Solutions to Exercise 3.6.6

1. Calculate the value of x :
 - (a) Label the sides of the triangle:



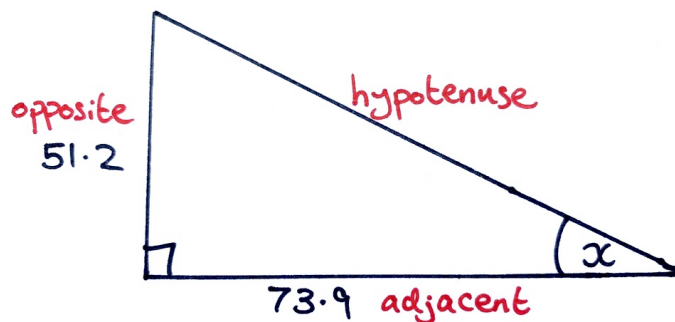
Using SOH-CAH-TOA:

$$\sin(x) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{19.4}{39.7}$$

Hence,

$$x = \sin^{-1}\left(\frac{19.4}{39.7}\right) = 29.25^\circ$$

(b) Label the sides of the triangle:



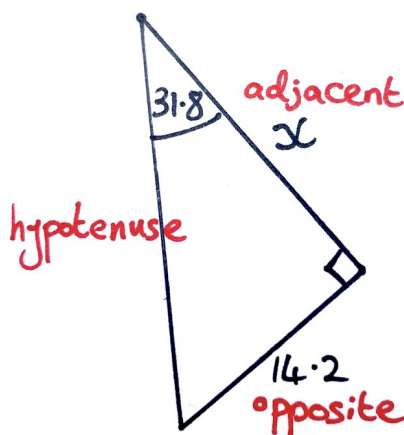
Then the two sides that we know are the opposite and the adjacent, so using SOH-CAH-TOA:

$$\tan(x) = \frac{\text{opposite}}{\text{adjacent}} = \frac{51.2}{73.9}$$

Hence, taking the inverse tan function to obtain x :

$$x = \tan^{-1}\left(\frac{51.2}{73.9}\right) = 34.72^\circ$$

(c) Label the sides of the triangle:

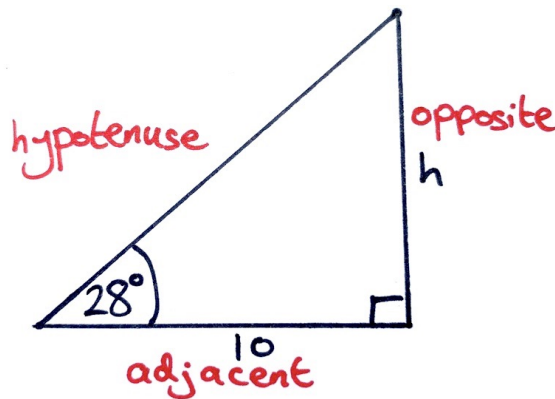


Using SOH-CAH-TOA:

$$\tan(31.8) = \frac{\text{opposite}}{\text{adjacent}} = \frac{14.2}{x}$$

$$\therefore x = \frac{14.2}{\tan(31.8)} = 22.90 \text{ cm}$$

2. Let the unknown height of the tree be h . Then, begin by sketching the situation and labelling the sides of the resulting right-angled triangle:



Thus, we know the value of the adjacent and need to determine the opposite. Applying SOH-CAH-TOA:

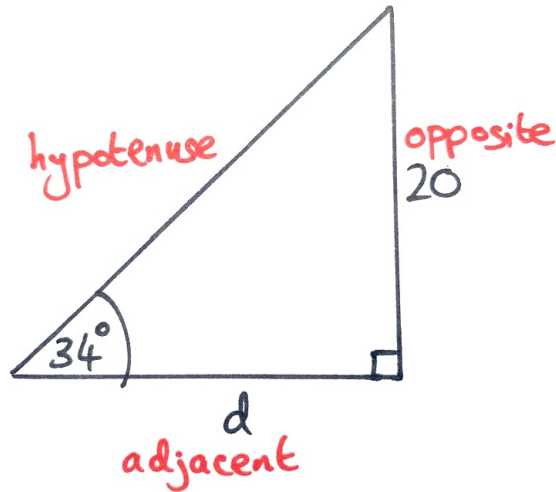
$$\tan(28) = \frac{\text{opposite}}{\text{adjacent}} = \frac{h}{10}$$

Re-arranging to solve for h :

$$h = 10 \times \tan(28) = 5.3 \text{ m}$$

3. Let the unknown horizontal distance be d .

Sketching the situation and labelling the sides of the resulting right-angled triangle:



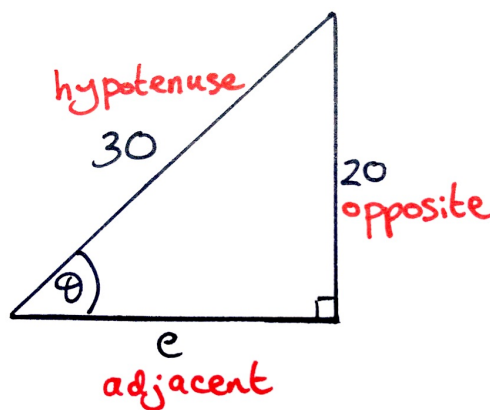
Using SOH-CAH-TOA:

$$\tan(34) = \frac{\text{opposite}}{\text{adjacent}} = \frac{20}{d}$$

Hence, solving for d :

$$d = \frac{20}{\tan(34)} = 29.65 \text{ m}$$

4. Let the angle between the cable and the ground be θ , and let the horizontal distance from it's anchor point to the foot of the vertical mast be e .



First, to find the angle θ we use the fact that we know both the opposite and the hypotenuse:

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{20}{30}$$

Taking inverse sine of both sides:

$$\theta = \sin^{-1}\left(\frac{2}{3}\right) = 41.8^\circ$$

Then to find the side e , which is the adjacent, use the fact that we know the hypotenuse and now the angle θ as well:

$$\cos(41.8) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{e}{30}$$

$$\therefore e = 30 \times \cos(41.8) = 22.4 \text{ m}$$

Note:

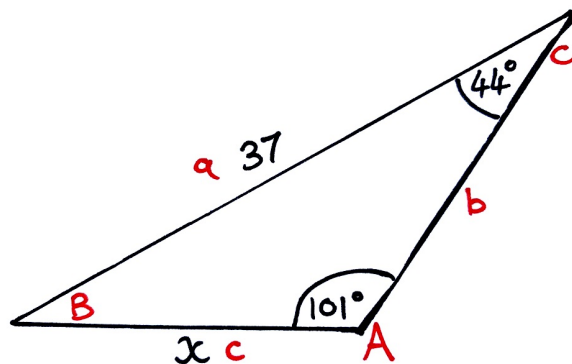
We often have several other paths to the solution, for example we could have used Pythagoras's theorem to find e without finding θ first:

$$30^2 = 20^2 + e^2$$

Transposing for e :

$$\begin{aligned}\therefore e &= \sqrt{30^2 - 20^2} \\ &= \sqrt{500} \\ &= 22.4 \text{ m}\end{aligned}$$

5. (a) Labelling the side-angle pairs in the triangle:



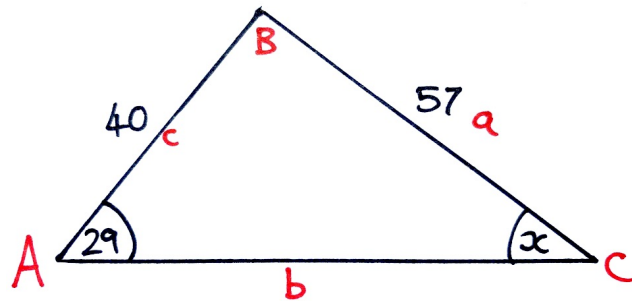
As we know one full side-angle pair (side of length 37 and angle 101°), and we know the angle (44°) opposite the side x that we need to find, we will use the sine rule. Substituting in the values from the triangle:

$$\frac{a}{\sin(A)} = \frac{c}{\sin(C)} \implies \frac{37}{\sin(101)} = \frac{x}{\sin(44)}$$

Then solving for x :

$$x = \sin(44) \times \frac{37}{\sin(101)} = 26.18 \text{ cm}$$

(b) Sketch the triangle and label the side-angle pairs:

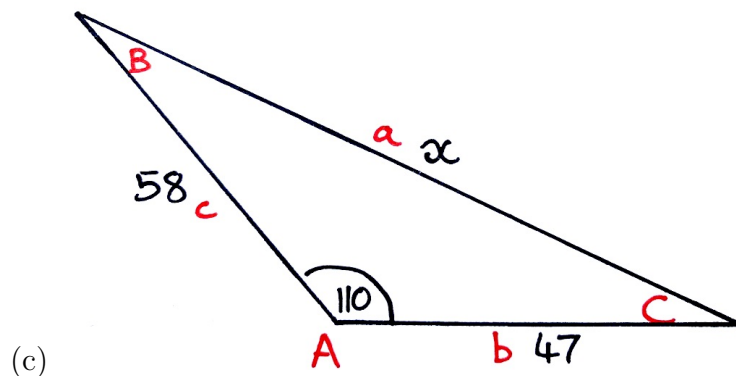


We know side (40) facing the angle x that we need to determine, and we know another full side-angle pair. Thus, we can use the sine rule:

$$\frac{\sin(A)}{a} = \frac{\sin(C)}{c} \implies \frac{\sin(29)}{57} = \frac{\sin(x)}{40}$$

Re-arranging to make x the subject:

$$\begin{aligned} \sin(x) &= 40 \times \frac{\sin(29)}{57} \\ \therefore x &= \sin^{-1}\left(\frac{40 \sin(29)}{57}\right) = 19.89^\circ \end{aligned}$$



In this case, we *don't* know any full side-angle pair so we cannot apply the sine rule. However, we know the other two sides and the angle between them which faces the side we need to find. Thus, using the cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

and substituting in our variables:

$$x^2 = 47^2 + 58^2 - 2 \times 47 \times 58 \times \cos(110) = 7437.693821$$

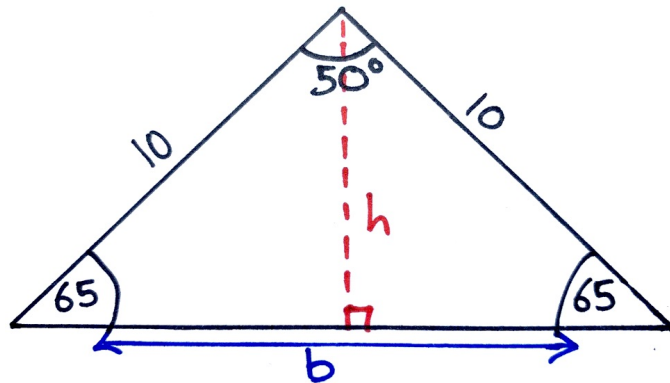
Hence, taking the square root of both sides:

$$x = 86.24 \text{ cm}$$

6. Since we know two of the three angles are 65° , the remaining angle is:

$$180 - (65 + 65) = 180 - 130 = 50^\circ$$

Sketching the isosceles triangle with the information given, and labelling the base b and height h :



Next, we find the perpendicular height h . To do this, consider one of the halves, which form right-angled triangles with known hypotenuse of 10 and where h is the opposite side to the known angle of 65° .

Thus, using SOH-CAH-TOA:

$$\sin(65) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{h}{10}$$

Transposing for h :

$$\begin{aligned}
 h &= 10 \times \sin(65) \\
 &= 9.0630779\dots \\
 &\approx 9.1 \text{ cm}
 \end{aligned}$$

Then to find the base b , use the fact that half of the base forms the adjacent of each of the right-angled triangles. Again applying SOH-CAH-TOA:

$$\cos(65) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{0.5 \times b}{10}$$

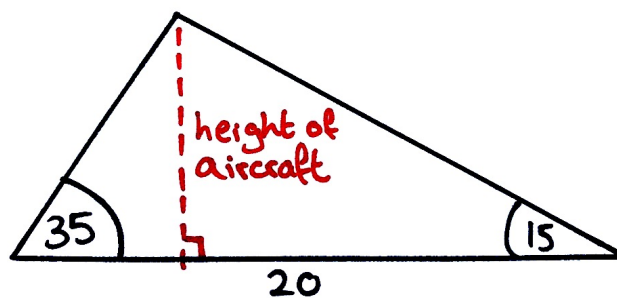
Hence:

$$\frac{1}{2}b = 10 \cos(65) = 4.22618\dots$$

And so:

$$\begin{aligned}
 b &= 8.452365\dots \\
 &\approx 8.5 \text{ cm}
 \end{aligned}$$

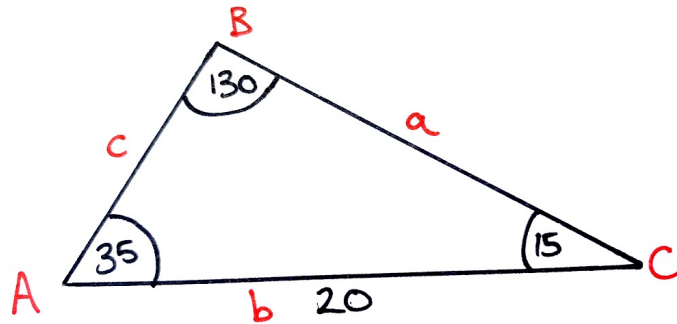
7. Begin by sketching an overview of the situation:



We are not able to calculate the altitude immediately. If we could find one of the other two sides, then we would know an angle and one side of one of the two right-angled triangles, and we could then use that information and *SOH – CAH – TOA* to determine the *opposite* side that is the altitude of the plane that we need.

Let's begin by determining the value of the missing angle:

$$180 - 35 - 15 = 130^\circ$$



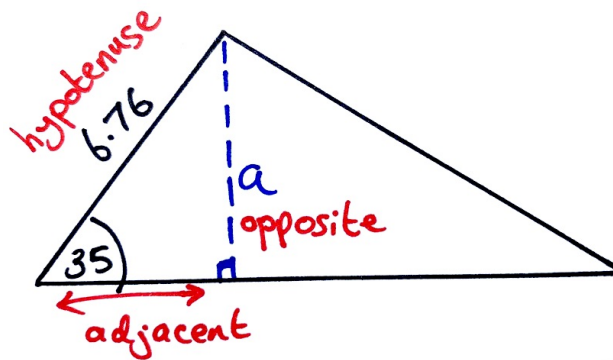
Now we have a complete side-angle pair, so we can apply the sine rule to calculate one of the other sides:

$$\frac{b}{\sin(B)} = \frac{c}{\sin(C)} \implies \frac{20}{\sin(130)} = \frac{c}{\sin(15)}$$

Re-arranging to solve for side c :

$$c = \sin(15) \times \frac{20}{\sin(130)} = 6.76 \text{ miles}$$

Label another sketch to emphasise the right-angled triangle and the key information we now have:



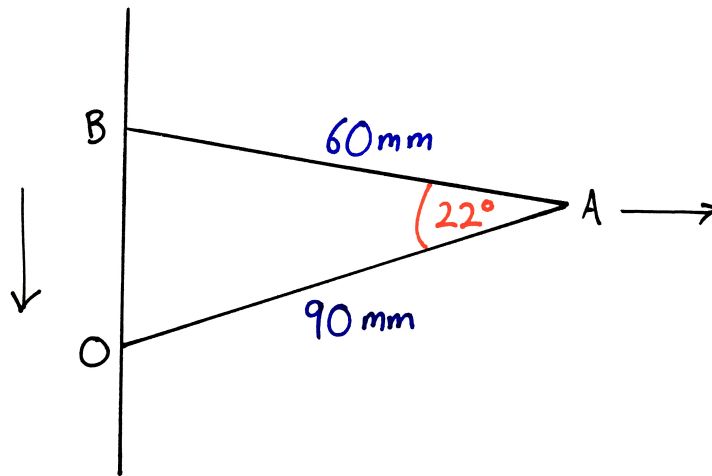
Finally we can calculate the altitude, which is the *opposite*, using SOH-CAH-TOA:

$$\sin(35) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{6.76}$$

Transposing for altitude a :

$$a = 6.76 \times \sin(35) = 3.88 \text{ miles}$$

8. As we want to find a length OB when we know the opposing angle and the other two sides, we can accomplish this using the cosine rule.



$$\begin{aligned}
 (OB)^2 &= (AB)^2 + (OA)^2 - 2(AB)(OA) \cos(22^\circ) \\
 &= 60^2 + 90^2 - 2(60)(90) \cos(22^\circ) \\
 &= 3600 + 8100 - 10800 \cos(22^\circ) \\
 &= 11700 - 10800(0.92718385) \\
 &= 1686.41437
 \end{aligned}$$

Taking the square root:

$$OB = \sqrt{1686.41437} = 41.0660$$

Hence, the length OB is 41.1mm (3 s.f.).